

Demand Estimation and Conduct in the Norwegian Cement Cartel

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1 Introduction

The empirical identification of market conduct in oligopolistic industries remains a central challenge in industrial organization. While the theoretical conditions under which collusion emerges and collapses are well understood, empirically distinguishing competitive from collusive behavior requires credible variation in the data and a well-specified structural model. This paper revisits the Norwegian cement cartel studied by Röller and Steen (2006), exploiting a unique institutional setting in which the same industry transitioned across three distinct market structures: a formal cartel with a capacity-sharing rule (1955–1968), a statutory monopoly following merger (1968–1982), and a subsequent period of competition.

Röller and Steen (2006) demonstrate that the cartel’s sharing rule generated systematic over-production incentives, ultimately rendering the merger welfare-improving relative to continued cartelization. Their analysis proceeds in two steps: estimating an inverse demand function via two-stage least squares, and recovering conduct parameters implied by the structural first-order conditions of each market regime. We replicate their empirical strategy and evaluate two dimensions of robustness that the original paper leaves unaddressed.

First, the original specification adopts an autoregressive distributed lag (ADL) model without formal justification of the lag order. We examine whether the chosen ADL(1,1) specification is adequate by conducting Breusch–Godfrey serial correlation tests and comparing information criteria across alternative lag lengths. Second, the original paper pools cartel and monopoly observations when estimating demand, implicitly assuming structural stability across regimes. We test this assumption by estimating an interaction specification that allows the demand slope to shift following the 1968 merger.

We find that neither extension improves upon the original specification given the available data. The lag-order diagnostics suggest that richer dynamic structures may be warranted, but the small sample renders them impractical due to overfitting and multicollinearity. The structural break test yields a statistically significant interaction term, but the implied post-merger elasticity is economically implausible, indicating identification failure rather than genuine structural

change. Accordingly, we retain the parsimonious ADL(1,1) pooled specification as the preferred baseline.

Building on this demand estimate, we extend the original paper by evaluating conduct parameters for both the cartel and monopoly periods under alternative identifying restrictions. Estimates for the cartel period suggest systematic deviation from the theoretical benchmark with optimal quantity allocation by the common sales office. Monopoly-period estimates are substantially less stable and sensitive to the limitations in sample size and cost structure constraint.

The remainder of the paper proceeds as follows. Section 2 describes the institutional background and presents the theoretical model. Section 3 reports demand estimation results. Section 4 evaluates conduct. Section 5 concludes.

2 Background

2.1 Norwegian Cement Cartel

The Norwegian cement industry operated as a formal cartel from 1955 until 1968, centralising all domestic sales through a common sales office (*Sementkontoret*). Under this arrangement, domestic revenues were allocated among member firms in proportion to their installed production capacity, creating a sharing rule that tied each firm’s income directly to its capacity share rather than to its actual domestic sales. Because firms could freely export any quantity at the prevailing world price—without rebate or penalty from the sales office—the sharing rule generated a systematic incentive to expand capacity and overproduce for export, even when the world price fell below marginal cost.¹ The persistence of these export losses ultimately rendered the cartel arrangement inefficient, and in 1968 the three incumbent firms merged into a single statutory monopoly.

¹Following Röller and Steen (2006), we abstract from import throughout, reflecting the negligible import penetration observed in the data.

2.2 Theoretical Model

2.2.1 Setup

Let there be N symmetric domestic cement producers (with $N = 3$ in the Norwegian case). Inverse domestic demand is linear:

$$P_t = a_t - bQ_t^D$$

where $Q_t^D = \sum_i q_{it}^D$ denotes aggregate domestic sales and a_t captures demand shifters. Each firm i operates with installed capacity K_{it} and allocates output between domestic sales q_{it}^D and exports q_{it}^X , subject to the capacity constraint

$$q_{it}^D + q_{it}^X \leq K_{it}.$$

The export market is competitive and the world price R_t is taken as given. The individual profit function is

$$\pi_{it} = P_t q_{it}^D + R_t q_{it}^X - c(q_{it}^D + q_{it}^X),$$

where c denotes constant marginal cost. We maintain this constant-marginal-cost assumption throughout, following the original paper, while noting in Section 4 that it may be a binding restriction for the monopoly period.

A key implication of the setup is that exports are loss-making whenever $R_t < c$. By calculation marginal costs, Röller and Steen (2006) confirms that it was the case throughout the cartel period.

2.2.2 Case 1: Cartel with Sharing Rule

Under the cartel, the common sales office maximises joint domestic profit, treating the world price R_t as the opportunity cost of each unit sold domestically rather than exported. The relevant first-order condition is derived in Section A.1 and takes the form

$$P'(Q_t^D)Q_t^D + P(Q_t^D) = R_t.$$

The right-hand side reflects the fact that the marginal unit of domestic sales displaces an export that would have earned R_t ; the world price therefore plays the role of marginal cost in the cartel's pricing rule. Under full coordination, $\lambda = 1$ in the conduct equation of Section 4.

2.2.3 Case 2: Monopoly

Following the 1968 merger, a single firm chooses domestic output to maximise profit. Since the capacity constraint is assumed non-binding and export is loss-making at equilibrium, the monopolist solves

$$\max_{Q_t^M} P(Q_t^M)Q_t^M - cQ_t^M.$$

The first-order condition is

$$P'(Q_t)Q_t + P(Q_t) = c.$$

This is the standard monopoly pricing rule, with marginal cost c on the right-hand side rather than the world price R_t .

2.2.4 Case 3: Symmetric Cournot Benchmark

As a competitive counterfactual, Röller and Steen (2006) consider a symmetric Cournot equilibrium among the $N = 3$ incumbent firms. Each firm i solves

$$\max_{q_{it}} P(Q_t)q_{it} - cq_{it},$$

taking rivals' output as given. Imposing symmetry ($q_{it} = \frac{Q_t}{N}$ at equilibrium) yields

$$\frac{1}{N}P'(Q_t)Q_t + P(Q_t) = c.$$

The three first-order conditions differ only in the coefficient on the markup term (1 for cartel and monopoly, $\frac{1}{N}$ for Cournot) and in whether the right-hand side is R_t or c . This nested structure motivates the conduct-parameter approach of Section 4, where we estimate a single λ that characterises the degree of market power exercised by the industry.

Summary statistics for the pre-merger, post-merger, and full samples are reported in [Table 1](#), [Table 2](#), and [Table 3](#) respectively. Production and domestic consumption increase substantially after the merger, consistent with the disappearance of the overproduction-for-export incentive under the cartel’s sharing rule. Input prices rise markedly in the post-merger period, reinforcing the importance of controlling for cost shifters when estimating demand.

3 Estimation of Demand

3.1 Specification

Given the homogeneity of cement, product differentiation is unlikely to play a meaningful role in demand. We therefore estimate a parsimonious inverse demand function following the specification of Röller and Steen (2006). The primary identification challenge is the endogeneity of domestic quantity: firms set output in response to price, so OLS estimates of the demand slope are inconsistent. We address this using cost shifters—the real wage index (w_t), the price of materials (p_t^m), and the price of electricity (p_t^e).

One thing to note is that Röller and Steen (2006) include the export price R_t as an additional instrument in their Stata code without explicit discussion. We exclude it from our baseline specification. Adding R_t to the instrument set reduces the first-stage F -statistic from 2.00 to 1.63 while leaving second-stage coefficients virtually unchanged ($\hat{\beta}_q = -0.000829$ vs. -0.000853) and improving the Sargan statistic ($p = 0.618$ vs. $p = 0.438$). The export price therefore contributes no meaningful identifying variation for domestic quantity conditional on the included regressors, while consuming an additional degree of freedom in the already weak first stage. We therefore retain the three cost shifters as the baseline instrument set.

To capture sluggish adjustment in the construction sector, we estimate an Autoregressive Distributed Lag specification. The baseline inverse demand equation is

$$p_{n,t} = \beta_0 + \beta_1 q_{n,t} + \beta_2 q_{n,t-1} + \alpha_1 p_{n,t-1} + \gamma_1 \text{byan}_t + \gamma_2 \text{byan}_{t-1} + \varepsilon_t,$$

where byan_t denotes the Norwegian building activity index. The lagged price term $p_{n,t-1}$ captures persistence in domestic pricing, while the lagged quantity term $q_{n,t-1}$ allows for distributed demand effects arising from construction planning horizons and contractual rigidities.

3.1.1 Lag Order Selection

A limitation of the original specification is that the ADL(1,1) lag structure is adopted without formal justification. We examine its adequacy using the Breusch–Godfrey (BG) serial correlation test alongside the Akaike and Bayesian information criteria across lag lengths one through six. Results are reported in [Table 5](#), [Table 6](#), and [Table 7](#).

For domestic prices and the building activity index, residual autocorrelation is not a concern at lag one: BG p -values exceed 0.10 across all specifications. For domestic consumption, however, the BG test strongly rejects the null of no serial correlation at lag one ($p = 0.01$), with the rejection disappearing from lag two onward. Moreover, both AIC and BIC decline monotonically as lag length increases across all three series, suggesting that richer dynamic structures could improve fit.

We nonetheless retain the ADL(1,1) specification as the preferred baseline. The first-stage VIF diagnostics reported in [Table 4](#) make clear why: byan lags carry VIF values of 61–128, and the two quantity lags jointly produce VIF values of 10–13, indicating near-perfect collinearity among the regressors of richer specifications. With only 28 usable observations, the degrees of freedom cost of extending the lag order outweighs the potential reduction in residual autocorrelation. The ADL(1,1) specification represents the best available balance between dynamic adequacy and estimation precision in this sample.

3.1.2 Structural Stability

The pooled specification implicitly assumes that the inverse demand relationship remained stable across the cartel and monopoly periods. This assumption is potentially restrictive: the 1968 merger eliminated the overproduction incentive, which may have altered the relationship between domestic quantity and price. We test for a structural break by estimating

$$p_{n,t} = \beta_0 + \beta_1 q_{n,t} + \beta_2 \text{post1968}_t + \beta_3 (q_{n,t} \times \text{post1968}_t) + \Gamma X_t + \varepsilon_t,$$

where X_t collects lagged prices, lagged quantities, and the building activity index.

The interaction term β_3 is statistically significant (Panel 5, [Table 8](#)), suggesting that the relationship between domestic quantity and price shifted after the merger. However, the implied post-merger demand slope ($\beta_1 + \beta_3$) is positive, which is economically implausible for a homogeneous-good market. This sign reversal does not reflect a genuine structural change; it reflects identification failure. The post-merger subsample contains only 15 observations, while the instrument set must be augmented for the interaction specification. The cross-period variation in instrument strength compounds the problem: pre-merger standard deviations of the real wage and material price index are 14,625 and 19.8 respectively, compared with 40,757 and 44.4 post-merger, so the instruments are not comparable in strength across the two subperiods. The pooled specification therefore remains the preferred baseline.

3.2 Results

All IV estimates are reported in [Table 8](#). The preferred ADL(1,1) specification (Panel 3) successfully replicates the coefficient reported by Röllner and Steen (2006), with minor difference coming from the exclusion of the instrument export price. The estimated quantity slope is negative and statistically significant ($\hat{\beta}_1 = -0.00085$, $p = 0.034$), consistent with a downward-sloping inverse demand function. The lagged price term is positive and marginally significant ($\hat{\alpha}_1 = 0.527$, $p = 0.060$), indicating substantial price persistence in the domestic cement market. The lagged quantity coefficient is positive and significant ($\hat{\beta}_2 = 0.00072$, $p = 0.041$), capturing the delayed response of domestic demand to quantity changes.

The Wu–Hausman test strongly rejects the null of exogenous quantity ($p = 0.002$), confirming that OLS estimation would yield inconsistent estimates and that IV identification is necessary. The Sargan overidentification test does not reject instrument validity ($p = 0.438$), providing no evidence against the exclusion restrictions. These diagnostics support the internal consistency of the specification.

The weak instrument test, however, raises a concern: the first-stage F -statistic for the excluded instruments is 2.00 ($p = 0.146$), falling well below the conventional threshold of 10. This indicates that the cost-shifter instruments have limited explanatory power for domestic quantity conditional on the included regressors, so finite-sample 2SLS estimates may be biased toward OLS and confidence intervals may understate true uncertainty. As a robustness check, we re-estimated the demand equation using LIML, which yielded a quantity coefficient of -0.00103 (s.e. 0.000486), approximately 21 percent larger in magnitude than the 2SLS estimate. The Anderson–Rubin confidence interval confirms that the negative demand slope is robust to weak instruments under both estimators. Weak identification is a recognised constraint of the Norwegian cement data, and we retain the ADL(1,1) IV estimates to maintain comparability with Röllér and Steen (2006) while acknowledging that the conduct results in Section 4 should be interpreted with this limitation in mind.

The richer dynamic specifications—ADL(2,2) and partial-lag alternatives (Panels 3–4)—do not resolve the weak instrument problem and introduce additional difficulties. As documented above, lagged regressors in the first stage carry VIF values as high as 128, and the ADL(2,2) specification (Panel 4) reduces the usable sample to 23 observations. The quantity coefficient loses significance and its sign becomes positive in Panel 4, a pattern consistent with severe multicollinearity rather than a genuine demand relationship. We therefore do not draw inference from these extended specifications.

4 Evaluation of Conduct

4.1 Specification

As established in Section 2, the first-order conditions of the cartel, monopoly, and Cournot models share the unified structure

$$\lambda \cdot P'(Q_t)Q_t + P(Q_t) = MC_t,$$

where $\lambda = 1$ and $MC_t = R_t$ under the cartel, $\lambda = 1$ and $MC_t = c$ under monopoly, and $\lambda = 1/3$ and $MC_t = c$ under symmetric Cournot competition. Estimating λ from data therefore allows us to assess how closely observed pricing behaviour conforms to each theoretical benchmark.

Because the inverse demand equation is estimated using an ADL(1,1) specification, the relevant demand slope entering the conduct equation is not uniquely defined. In a static setting, the conduct parameter is recovered using the contemporaneous slope β_1 . In a dynamic setting, however, a change in current quantity affects not only the contemporaneous price but also future prices through the persistence of lagged price and quantity terms. The estimated inverse demand equation implies two candidate slopes. The short-run slope captures the immediate marginal effect of quantity on price:

$$\beta_{\text{sr}} = \beta_1.$$

The long-run slope incorporates the full dynamic propagation through lagged adjustments:

$$\beta_{\text{lr}} = \frac{\beta_1 + \beta_2}{1 - \rho},$$

where ρ is the estimated coefficient on the lagged price term.

The choice between β_{sr} and β_{lr} has a natural economic interpretation in the cement industry. Cement procurement is governed predominantly by long-term construction contracts, so pricing decisions are more likely to reflect the cumulative effect of a quantity change over time than its instantaneous impact alone. On this reasoning, β_{lr} is arguably the more relevant slope for conduct estimation, and the short-run estimate should be treated as a lower bound. We nonetheless report both as robustness checks. The markup terms entering the conduct regressions are defined as

$$\mu_{\text{sr}} = -Q_t \cdot \beta_{\text{sr}}, \quad \mu_{\text{lr}} = -Q_t \cdot \beta_{\text{lr}}.$$

4.2 Estimation Strategy

We estimate the conduct parameter λ under three alternative identifying restrictions, reported in the appendix tables. The specifications are nested and progressively impose the theoretical restrictions implied by the structural model.

Model 1: Unrestricted specification. No restrictions are imposed on either the intercept or the coefficient on the opportunity-cost proxy:

$$P_t = \alpha + \delta MC_t + \lambda(-Q_t\beta) + \varepsilon_t.$$

This specification allows both the level and the sensitivity to the proxy variable to vary freely, providing a flexible reduced-form benchmark. The estimated λ should not be interpreted as a structural conduct parameter in the conventional industrial organization sense. Instead, it measures the extent to which the observed pricing relationship conforms to the first-order condition implied by the office model.

Model 2: Coefficient-constrained specification. The theoretical restriction that the proxy enters with coefficient one is imposed:

$$P_t - MC_t = \alpha + \lambda(-Q_t\beta) + \varepsilon_t.$$

This follows more closely the first-order condition implied by the model. Under the theoretical benchmark $\lambda = 1$; values below unity indicate that observed pricing deviates from the profit-maximizing level implied by the structural model.

Model 3: Intercept- and coefficient-constrained specification. A zero intercept is additionally imposed:

$$P_t - MC_t = \lambda(-Q_t\beta) + \varepsilon_t.$$

This specification most closely adheres to the theoretical pricing condition. The estimated λ can therefore be interpreted as the degree to which observed pricing conforms to the benchmark first-order condition, with no residual pricing wedge allowed.

4.3 Results

Results for the cartel period are reported in [Table 9](#), [Table 10](#), and [Table 11](#), and monopoly-period estimates in [Table 12](#) and [Table 13](#).

4.3.1 Cartel Period

Specifications using the export price as the opportunity-cost proxy (Table 9) yield positive and statistically significant conduct estimates under both constrained models. Under the coefficient-constrained specification (Columns 3–4), $\hat{\lambda}$ ranges from 0.08 using the short-run slope to 0.24 using the long-run slope. The fully constrained specification (Columns 5–6) produces similar estimates of 0.09 and 0.25. Results using the authors’ predicted marginal cost as the proxy (Table 10) are broadly consistent, with long-run estimates of 0.11 and 0.14 under the two constrained models. Across all three proxies and both constrained specifications, $\hat{\lambda}$ is positive, statistically significant, and substantially below the theoretical benchmark of one. The unrestricted specification (Columns 1–2) produces negative and insignificant estimates throughout, indicating substantial sensitivity to the intercept restriction.

These estimates uniformly indicate that $\hat{\lambda} < 1$. Given that the long-run demand slope is the more appropriate choice for the cement industry, the long-run estimates of $\hat{\lambda} \approx 0.24$ –0.25 are our preferred figures for the cartel period. These imply that the common sales office set domestic quantity above the profit maximizing level.

Two potential bias are worth considering. First, the constant marginal cost assumption maintained throughout may be misspecified: if the true cost curve is upward-sloping, the theoretical markup implied by R_t overstates the actual profit-maximising markup, and $\hat{\lambda} < 1$ partly reflects this misspecification rather than genuine quantity distortion. Second, the export price R_t is measured with error as a proxy for world price, which is the true opportunity cost of domestic supply. This measurement error would attenuate the estimated conduct parameter toward zero. Both sources of downward bias suggest that the true degree of coordination may be higher than the point estimates imply.

4.3.2 Monopoly Period

Monopoly-period conduct estimates are considerably less stable. Using export price as a proxy (Table 12), the coefficient-constrained specification yields $\hat{\lambda} \approx 0.24$ –0.71 depending on

the slope choice, while the fully constrained model produces long-run estimates close to unity. However, the unrestricted specification yields economically implausible and statistically insignificant results, and estimates based on short-run average cost ([Table 13](#)) vary substantially across specifications, including sign reversals under the coefficient-constrained model.

The instability of monopoly-period estimates reflects two compounding limitations. First, the post-merger sample contains only 15 observations, providing limited degrees of freedom for identification. Second, the constant marginal cost assumption is also potentially more restrictive for the monopoly period: the merged entity operated a larger and more complex production network, making a flat cost curve less plausible than during the cartel era. Together, these factors preclude reliable inference on monopoly-period conduct, and we do not draw strong conclusions from these estimates.

4.3.3 Summary

Taken together, the conduct estimates support the view that the cartel’s common sales office exercised substantial but incomplete market power. The preferred long-run estimates for the cartel period consistently indicate $\hat{\lambda} < 1$, which we partially attribute to model misspecification under constant marginal costs and attenuation bias from measurement error in the opportunity-cost proxy, rather than to any fundamental failure of coordination. Monopoly-period conduct cannot be reliably identified given the available data.

5 Conclusion

We replicate the demand and conduct analysis of Röller and Steen (2006) using data from the Norwegian cement industry and extend the original paper along three dimensions: a formal evaluation of the ADL lag structure, a test for structural instability around the 1968 merger, and an explicit estimation of conduct parameters for both the cartel and monopoly periods.

On the demand side, neither extension improves upon the original specification given the available data. Although diagnostics suggest that richer dynamic structures could improve the

model in theory, the small sample and strong collinearity among lagged variables make extended specifications unstable.

On the conduct side, the results indicate significant but incomplete market power during the cartel period. Estimates remain well below the theoretical monopoly benchmark, a gap we attribute primarily to downward bias from two sources: misspecification of the cost structure; and attenuation from measurement error in the proxy for the opportunity cost of domestic supply. Despite these limitations, we conclude that Norwegian cement cartel appears to have exercised significant pricing power, whereas Monopoly-period conduct cannot be reliably identified.

For future work, richer cost specification would sharpen conduct identification for the monopoly period. Additional instruments with stronger first-stage relevance would address the weak identification problem that limits inference throughout. Finally, the structural break specification, while infeasible here due to sample limitations, could be revisited if data with longer time series or higher frequency is available.

A Appendix

A.1 Mathematical appendix

A.1.1 Cartel with sharing rule

Norway Cement Cartel had a sharing rule based on capacity. The share of profit of a firm s_{it} was decided by a following rule:

$$s_{it} = \frac{K_{it}}{\sum_j K_{jt}}$$

with K_{it} capacity of firm i . Excess domestic stock is exported at price R_t , that is $q_{it}^X = K_{it} - q_{it}^D$. Profit function of firm i writes as:

$$\pi_{it} = P_t s_{it} Q_t^D + R_t (K_{it} - q_{it}^D) - c K_{it}$$

As in the original paper, we proceed first by determining optimal aggregate quantity Q_t^D then idiosyncratic quantity q_{it} . It results that:

$$\max_{Q_t^D} \Pi_t = P_t(Q_t^D) Q_t^D + R_t \left(\sum_i K_{it} - Q_t^D \right) - c \sum_i K_{it}$$

We solve by backward induction. Given an optimal value of Q^D , we have²:

$$\pi_i = P \frac{K_i}{K_i + \sum_{i \neq j} K_j} Q^D + R(K_i - s_i Q^D) - c K_i$$

Differentiating w.r.t K_i ,

$$\begin{aligned} (P - R) \frac{1}{Q} \left(1 - \frac{K_i}{Q} \right) Q^D + (R - c) &= 0 \\ \Leftrightarrow (1 - s_i) \frac{Q^D}{Q} + R - c &= 0 \end{aligned}$$

One can then derive the optimal collective chosen domestic production:

$$P'(Q^D) Q^D + P(Q^D) = R$$

A.1.2 Monopoly

The merged monopolist chooses Q_t^M to maximise total profit. Since the capacity constraint is non-binding and exporting is loss-making ($R_t < c$) at the optimum, the problem reduces to

$$\max_{Q_t^M} P(Q_t^M) Q_t^M - c Q_t^M.$$

The first-order condition is

$$P'(Q_t^M) Q_t^M + P(Q_t^M) - c = 0$$

²We omit the time index for clarity.

$$\therefore P'(Q_t^M)Q_t^M + P(Q_t^M) = c.$$

A.1.3 Benchmark Symmetric Cournot

With $N = 3$ symmetric firms, firm i solves

$$\max_{q_{it}} P(Q_t)q_{it} - cq_{it}, \quad Q_t = q_{it} + Q_{-it}.$$

The first-order condition for firm i is

$$P'(Q_t)q_{it} + P(Q_t) = c.$$

Imposing symmetry, $q_{it} = \frac{Q_t}{N}$, gives

$$\frac{1}{N}P'(Q_t)Q_t + P(Q_t) = c.$$

Comparing the three conditions reveals a unified structure:

$$\lambda \cdot P'(Q_t)Q_t + P(Q_t) = MC_t$$

where $\lambda = 1$ and $MC_t = R_t$ under the cartel, $\lambda = 1$ and $MC_t = c$ under monopoly, and $\lambda = \frac{1}{N}$ and $MC_t = c$ under Cournot competition. This nesting motivates the conduct-parameter estimation strategy in Section 4.

A.2 Estimation of demand and conduct

Table 1. Summary statistics for the pre-merger period.

Variable	Mean	SD	Min	Max
Total production	1397665	452575.31	799878	2297088
Domestic price	459.49	72.18	359.08	594.76
Total exports	227745.33	247054.1	800	840596
Export price	393.92	107.1	235.07	641.16
Price of materials	99.23	19.31	63.37	121.73
Price of electricity	100.66	49.06	50.06	185.82
Real wage	96337.91	15128.46	69173	116776.6
Domestic consumption	1128108.58	232149.24	740249	1539344

Table 2. Summary statistics for the post-merger period.

Variable	Mean	SD	Min	Max
Total production	2425886.86	328250.14	1786000	2740169
Domestic price	593.37	90.36	409.72	706.29
Total exports	538526.33	456582.76	0	1217277
Export price	255.99	38.99	227.54	373.19
Price of materials	153.24	42.65	71.96	208.38
Price of electric- ity	99.88	35.84	50.82	150.23
Real wage	167627.40	39990.56	115812.92	211595.31
Domestic con- sumption	1631846.86	110257.77	1403078	1795089

Table 3. Summary statistics for the full sample.

Variable	Mean	SD	Min	Max
Total production	1911775.93	651611.24	799878	2740169
Domestic price	524.13	105.03	359.08	706.29
Total exports	409034.25	409493.55	0	1217277
Export price	327.33	106.59	227.54	641.16
Price of materials	125.30	42.25	63.37	208.38
Price of electric- ity	100.28	42.43	50.06	185.82
Real wage	130753.52	46597.23	69173	211595.31
Domestic con- sumption	1371292.57	313395.02	740249	1795089

Table 4. Variance inflation factors from the ADL(2,2) first-stage regression for domestic quantity.

Variable	VIF
byan	75.75
pnr_lag1	7.70
pnr_lag2	9.10
qn_lag1	9.86
qn_lag2	13.09
byan_lag1	128.35
byan_lag2	61.45
wayr	13.49
pmr	7.72
per	6.15

Note. VIF values above 10 indicate problematic collinearity; values above 30 indicate near-perfect collinearity.

Table 5. Table of lag order autocorrelation for price.

Lag order	AIC	BIC	BG p-value	adj. R^2
1	311.14	315.14	0.14	0.7
2	300.91	306.09	0.47	0.71
3	291.8	298.09	0.24	0.71
4	282.25	289.56	0.67	0.71
5	272.12	280.37	0.15	0.72
6	260.53	269.62	0.3	0.75

Table 6. Table of lag order autocorrelation for demand index.

Lag order	AIC	BIC	BG p-value	adj. R^2
1	317.22	321.22	0.95	0.98
2	308.73	313.92	0.47	0.97
3	298.56	304.85	0.42	0.97
4	286.76	294.07	0.66	0.97
5	278.49	286.73	0.3	0.97
6	267.42	276.51	0.24	0.97

Table 7. Table of lag order autocorrelation for consumption.

Lag order	AIC	BIC	BG p-value	adj. R^2
1	725.09	729.09	0.01	0.9
2	693.53	698.72	0.42	0.91
3	670.65	676.94	0.58	0.89
4	646.64	653.96	0.26	0.87
5	623.17	631.42	0.58	0.84
6	598.53	607.61	0.42	0.82

Table 8. Two-Stage Least Squares regression table results of inverse demand

	<i>Dependent variable:</i>				
	pnr				
	(1)	(2)	(3)	(4)	(5)
qn	−0.0003*** (0.0001)	−0.001** (0.0004)	−0.001** (0.0004)	0.0001 (0.0003)	−0.001*** (0.0002)
post1968					−934.966** (430.492)
byan	0.373*** (0.065)	−0.148 (0.369)	−0.190 (0.297)	0.060 (0.236)	0.098 (0.198)
pnr_lag			0.532* (0.259)	0.305 (0.292)	
pnr_lag2				0.480 (0.376)	
qn_lag		0.001* (0.0004)	0.001** (0.0003)	0.0001 (0.0002)	0.0004** (0.0002)
byan_lag		0.482 (0.363)	0.334 (0.300)		0.076 (0.205)
qn_lag2				0.00003 (0.0002)	
byan_lag1				−0.301 (0.334)	
byan_lag2				0.203 (0.260)	
qn:post1968					0.001** (0.0003)
Constant	388.009*** (64.617)	451.367*** (109.109)	239.361* (135.189)	−180.004 (182.022)	741.716*** (101.969)
Observations	29	28	28	23	28
R^2	0.570	0.232	0.527	0.843	0.800
Adjusted R^2	0.537	0.098	0.420	0.753	0.743
Residual Std. Error	71.441 (df = 26)	101.552 (df = 23)	81.450 (df = 22)	58.303 (df = 14)	54.212 (df = 21)

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Table 9. Estimation of conduct parameter in cartel era, using export price proxy

	<i>Dependent variable:</i>					
	pnr		gap_pexpr		pnr (no const.)	
	(1)	(2)	(3)	(4)	(5)	(6)
pexpr	0.339 (0.270)	0.339 (0.270)			0.983 *** (0.067)	0.983 *** (0.067)
markup_sr	-0.218 (0.127)		0.080 *** (0.009)		0.086 *** (0.026)	
markup_lr		-0.640 (0.371)		0.235 *** (0.027)		0.252 *** (0.075)
Constant	543.517 ** (222.492)	543.517 ** (222.492)				
Observations	14	14	14	14	14	14
R^2	0.854	0.854	0.851	0.851	0.995	0.995
Adjusted R^2	0.827	0.827	0.839	0.839	0.994	0.994
Residual Std. Error	30.436 (df = 11)	30.430 (df = 11)	34.867 (df = 13)	34.867 (df = 13)	36.192 (df = 12)	36.192 (df = 12)
F Statistic	32.063 ***	32.063 ***	74.033 ***	74.033 ***	1,129.367 ***	1,129.367 ***
<i>Note:</i>				* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$		

Table 10. Estimation of conduct parameter in cartel era, using authors' predicted marginal cost as proxy

	<i>Dependent variable:</i>					
	pnr		gap_pmc		pnr (no const.)	
	(1)	(2)	(3)	(4)	(5)	(6)
predict- edmc	0.941 ***	0.941 ***			0.968 ***	0.968 ***
	(0.172)	(0.172)			(0.029)	(0.029)
markup_sr	0.037		0.036 ***		0.049 ***	
	(0.078)		(0.004)		(0.012)	
markup_lr		0.107		0.106 ***		0.143 ***
		(0.229)		(0.013)		(0.036)
Constant	23.580	23.580				
	(147.228)	(147.228)				
Observa- tions	14	14	14	14	14	14
R^2	0.955	0.955	0.842	0.842	0.999	0.999
Adjusted R^2	0.947	0.947	0.829	0.829	0.999	0.999
Residual	16.871	16.871	16.315	16.315	16.171	16.171
Std. Error	(df = 11)	(df = 11)	(df = 13)	(df = 13)	(df = 12)	(df = 12)
F Statistic	116.760 ***	116.760 ***	69.085 ***	69.085 ***	5,680.915 ***	5,680.915 ***
<i>Note:</i>				* $p < 0.1$; * * $p < 0.05$; * * * $p < 0.01$		

Table 11. Estimation of conduct parameter in cartel era, using short-run average cost as proxy

	<i>Dependent variable:</i>					
	pnr		gap_sac		pnr (no const.)	
	(1)	(2)	(3)	(4)	(5)	(6)
shortAC	0.103 *** (0.032)	0.103 *** (0.032)			0.175 *** (0.009)	0.175 *** (0.009)
markup_sr	−0.045 (0.105)		−0.920 *** (0.188)		0.199 *** (0.014)	
markup_lr		−0.132 (0.309)		−2.697 *** (0.551)		0.583 *** (0.041)
Constant	347.685 ** (149.077)	347.685 ** (149.077)				
Observations	14	14	14	14	14	14
R^2	0.914	0.914	0.648	0.648	0.997	0.997
Adjusted R^2	0.899	0.899	0.621	0.621	0.997	0.997
Residual Std. Error	23.282 (df = 11)	23.282 (df = 11)	704.455 (df = 13)	704.455 (df = 13)	27.250 (df = 12)	27.250 (df = 12)
F Statistic	58.695 ***	58.695 ***	23.982 ***	23.982 ***	1,996.684 ***	1,996.684 ***
<i>Note:</i>				* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$		

Table 12. Estimation of conduct parameter in monopoly era, using export price as proxy

	<i>Dependent variable:</i>					
	pnr	gap_pexpr	gap_pexpr	pnr		
	(1)	(2)	(3)	(4)	(5)	(6)
pexpr	0.148 (0.756)	0.148 (0.756)			0.134 (0.732)	0.134 (0.732)
markup_sr	0.256 (0.313)		0.242 *** (0.018)		0.401 ** (0.136)	
markup_lr		0.750 (0.918)		0.708 *** (0.052)		1.175 ** (0.398)
Constant	199.208 (385.442)	199.208 (385.442)				
Observations	14	14	14	14	14	14
R^2	0.091	0.091	0.935	0.935	0.980	0.980
Adjusted R^2	−0.074	−0.074	0.930	0.930	0.977	0.977
Residual	93.647	93.647	92.118	92.118	90.742	90.742
Std. Error	(df = 11)	(df = 11)	(df = 13)	(df = 13)	(df = 12)	(df = 12)
F Statistic	0.552 (df = 2; 11)	0.552 (df = 2; 11)	187.587 *** (df = 1; 13)	187.587 *** (df = 1; 13)	299.766 *** (df = 2; 12)	299.766 *** (df = 2; 12)
<i>Note:</i>				* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$		

Table 13. Estimation of conduct parameter in monopoly era, using short-run average cost as proxy

	<i>Dependent variable:</i>					
	pnr		gap_sac	gap_sac	pnr	
	(1)	(2)	(3)	(4)	(5)	(6)
shortAC	−0.014 (0.228)	−0.014 (0.228)			0.015 (0.214)	0.015 (0.214)
markup_sr	0.288 (0.281)		−0.099 *** (0.028)		0.418 *** (0.114)	
markup_lr		0.844 (0.823)		−0.289 *** (0.081)		1.224 *** (0.333)
Constant	202.494 (398.043)	202.494 (398.043)				
Observations	14	14	14	14	14	14
R^2	0.088	0.088	0.493	0.493	0.980	0.980
Adjusted R^2	−0.077	−0.077	0.454	0.454	0.977	0.977
Residual Std. Error	93.793 (df = 11)	93.793 (df = 11)	145.127 (df = 13)	145.127 (df = 13)	90.851 (df = 12)	90.851 (df = 12)
F Statistic	0.533 (df = 2; 11)	0.533 (df = 2; 11)	12.622 *** (df = 1; 13)	12.622 *** (df = 1; 13)	299.036 *** (df = 2; 12)	299.036 *** (df = 2; 12)
<i>Note:</i>				* $p < 0.1$; * * $p < 0.05$; * * * $p < 0.01$		